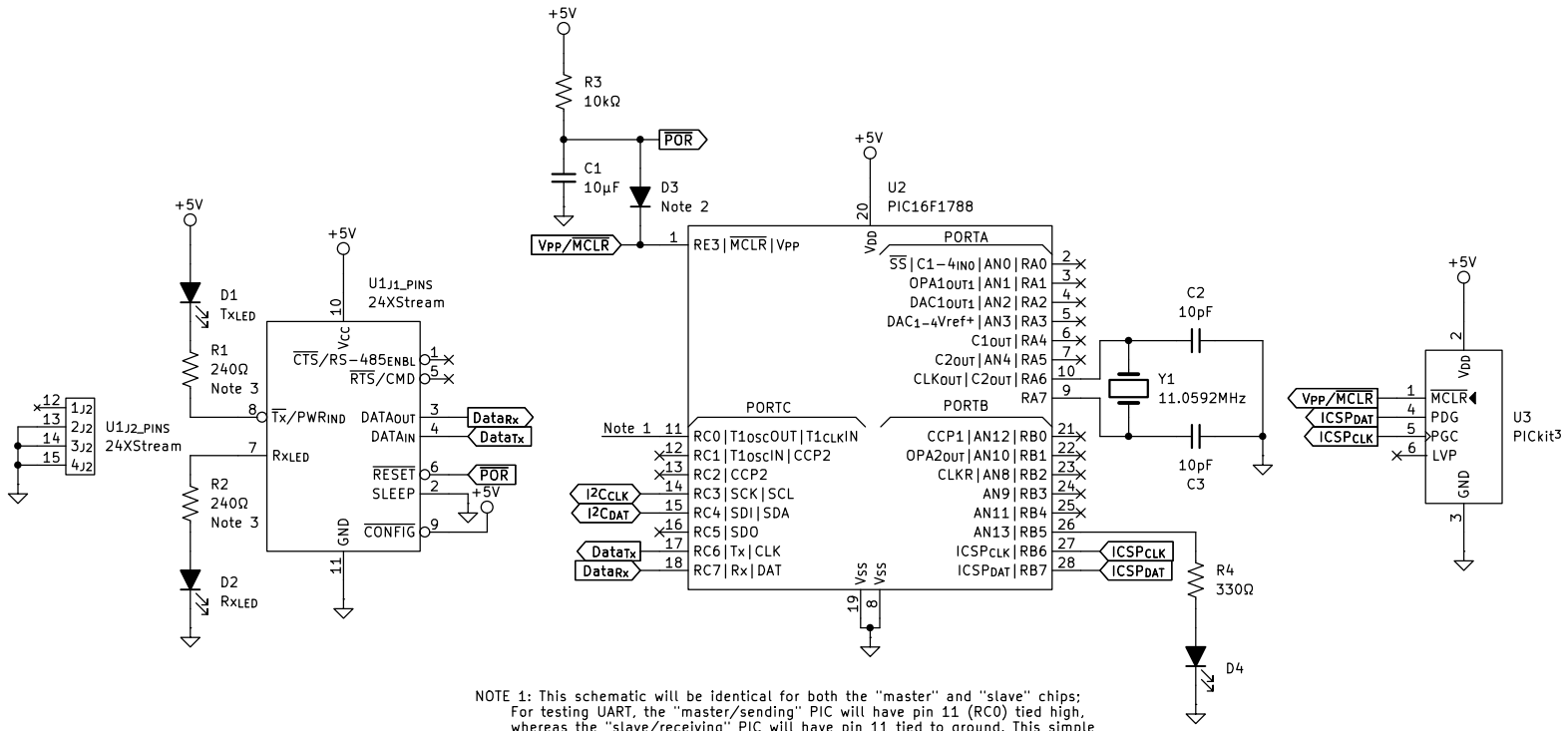


NOTE 1: This schematic will be identical for both the "master" and "slave" chips; For testing UART, the "master/sending" PIC will have pin 11 (RC0) tied high, whereas the "slave/receiving" PIC will have pin 11 tied to ground. This simple differentiation allows the function of both PICs to be tested without any software changes necessary.

NOTE 2: D3 was tested with a 1N4148 general purpose (GP) diode. Any equivelant GP diode should be sufficient.

NOTE 3: Datasheet rates $V_{OL_{MAX}}$=0.7V and I_{OL}/I_{OH}=±20mA_{MAX}
Assuming the LED drops 2V, VR_{LED}=2.3V.
R_{LED} = 2.3V/20mA_{MAX} = 115Ω$_{MIN}$
I_{LED} = 2.3V/240Ω = 9.583mA ∴ 50% of I_{MAX}

1132 Martin Luther King Jr. Way
Pocatello, ID 83209
Drawn By: Jonas Cram
Idaho State University

Idaho State University

ROBOTICS & COMMUNICATION SYSTEMS ENGINEERING TECHNOLOGY

Sheet: /
File: ControlStationTestCircuit.kicad_sch

Project: ControlStationTestCircuit

Title: UART Test Circuit

Size: USLetter	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1